



Third Semester B.Tech (AI&DS/AI&ML/CCE/CSE/ISE/R&AI) (CBCS) Degree Examinations

MA2001-1 – STATISTICS AND PROBABILITY THEORY

Duration: 3 Hours

Max. Marks:100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

2) **Statistical Table** may be provided.

20 Marks

- 1-

10. If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2 , then the limiting form of the distribution of $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ as $n \rightarrow \infty$ is _____
- A) Standard normal distribution
B) t-distribution
C) Chi-square distribution
D) Exponential distribution
11. In a binomial distribution $b(x, n, p)$, which of the following are parameters of the distribution
- A) x
B) n, p
C) λ
D) σ
12. When sample size $n < 30$ we use _____ distribution
- A) t-distribution
B) Standard normal
C) Chi-square distribution
D) Exponential distribution
13. Which of the following statements is not correct
- A) If X and Y are independent random variables then $\rho_{XY} = 0$.
B) If X and Y are independent random variables then $Cov(X, Y) = 0$.
C) Equation for the regression line of y on x is $y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$
D) If $Cov(X, Y) = 0$, then X and Y are independent
14. The value of $(\bar{x}, \bar{y})$ is _____ if the regression lines are $x = 0.7y + 5.2$ and $y = 0.3x + 2.8$
- A) (9, 5.52)
B) (0.7, 0.3)
C) (5.2, 2.8)
D) (1, 1)
15. Which statement is an example for uncorrelated random variables?
- A) Volume of a cube $V = L^3$
B) Rainfall and crop yield
C) Two coins being tossed simultaneously
D) Poverty and Crime
16. Which statement is not correct?
- A) Coefficient of correlation is the geometric mean between the regression coefficients
B) Coefficient of correlation is the arithmetic mean between the regression coefficients
C) Coefficient of correlation ρ lies in $[-1, 1]$.
D) Coefficient of correlation ρ is independent of origin.
17. Which among the following vectors is a probability vector?
- A) (0.25, -0.25, 0.5, 0.5)
B) (1/12, 1/2, 1/6, 0, 1/4)
C) (5/2, 0, 8/3, 1/6, 1/6)
D) (1/3, 0, -2/3, 1/5, 3/5)
18. Which matrix is stochastic matrix?
- A) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$
B) $\begin{pmatrix} 1/4 & 3/4 \\ 0 & 1 \end{pmatrix}$
C) $\begin{pmatrix} 1/2 & 1/3 \\ 0 & 1/2 \end{pmatrix}$
D) $\begin{pmatrix} 1/2 & 1/3 \\ -1 & 1/2 \end{pmatrix}$
19. Customer behavior where a customer moves from one queue to another in multiple channel situation is _____
- A) Balking
B) Reneging
C) Jockeying
D) Alternating
20. A customer who leaves the queue by losing his patience to wait is said to be
- A) Balking
B) Reneging
C) Jockeying
D) Alternating

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

1. a) The probability distribution of a random variable is given by the following table:

x	-2	-1	0	1	2	3
p(x)	0.1	k	0.2	2k	0.3	k

Find: i) the value of k, ii) $P(X > -1)$, iii) mean and variance of X.

6 L2 1 1

- b) Two random variables X and Y have joint p.d.f.

$$f(x, y) = \begin{cases} \frac{1}{3}(4 - x - y), & 1 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Compute E(X), E(Y) and E(XY).

6 L3 1 1

- c) In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for:

i) less than 10 minutes ii) between 10 and 12 minutes?

4 L2 2 1

2. a) A coin is known to come up heads 3 times as often as tails. This coin is tossed three times. Let X be the number of heads that appear. Write the probability distribution of X.

6 L3 1 1

- b) Suppose that the following table represents the joint p.d.f.

X \ Y	1	2	3
1	1/12	1/6	0
2	0	1/9	1/5
3	1/18	1/4	2/15

(i) Evaluate the marginal distribution functions of X and Y.

(ii) Obtain E (X) and E(XY).

6 L2 1 1

- c) The probability of a man hitting a target is 1/3. How many times must he fire so that the probability of his hitting the target at least once is more than 80%?

4 L3 2 1

3. a) Obtain the mean and variance of the Poisson distribution.
- b) In an examination 7% of the students score less than 35 marks and 89% of the students score less than 60 marks. Find the mean and standard deviation of the marks which is normally distributed. Values of z from standard normal table, corresponding to the probabilities 0.07, 0.89 and 0.93 are respectively -1.48, 1.23 and 1.48. Select suitable values from these values.
- c) In a certain college 40% of boys and 20% of girls are studying mathematics. The girls constitute 60% of the student body. If a student is selected at random and is found to be studying mathematics, find the probability that the student is a girl?

6 L2 2 1

6 L3 2 1

4 L3 1 1

Unit – II

4. a) State Central Limit Theorem. Apply it to find the size of the random sample selected from a normal distribution with mean μ and variance 100 such that $P[\mu - 5 < \bar{X} < \mu + 5] = 0.954$.

6 L3 3 1

- b) Fit a quadratic curve to the following data in the least square sense.

x	5	10	15	20	25
y	21	24	33	42	38

6 L2 4 1

- c) A random sample of size 100 is taken from a population having $\sigma = 5.1$ and sample mean $\bar{X} = 21.6$. Find 95% confidence interval for population mean μ .

4 L3 3 1

5. a) If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form $P = a + bW$ connecting P and W, using the following data, where P and W are taken in kg-wt. Compute P when $W = 150$ kg-wt.

P	12	15	21	25
W	50	70	100	120

6 L2 4 1

- b) Let a random sample of size 17 from a normal distribution $N(\mu, \sigma^2)$ yield $\bar{X} = 4.7, S^2 = 5.76$. Determine a 90% confidence interval for μ and σ^2 .

6 L3 3 1

- c) The equations of two regression lines are $x = 0.7y + 52$ and $y = 0.3x + 2.8$. Find the means of the variables and the correlation coefficient.

4 L2 4 1

6. a) Find the coefficient of correlation between industrial production and export using the following data.

Production (in crores of tons)	55	58	58	59	60
Export (in crores of tons)	37	41	37	40	45

6 L3 4 1

- b) Derive the formula: $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$

6 L3 4 1

- c) Show that the sample variance $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ is an unbiased estimator for population variance.

4 L2 3 1

Unit – III

7. a) A student's study habit is as follows:
If he studies one night, he is 60% sure not to study the next night. On the other hand if he does not study one night, he is 80% sure to study the next night. In the long run how often does he study?

6 L2 5 1

- b) Consider a Markov Chain with three states $S = \{1, 2, 3\}$ and transition matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

- i) Draw the state transition diagram for this chain.

- ii) If we know $P(X_1 = 1) = P(X_1 = 2) = 1/4$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

6 L3 5 1

- c) Find a fixed probability vector of the regular stochastic matrix:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

4 L2 5 1

8. a) Define probability vector and a stochastic matrix. If $A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$ is a stochastic matrix and $V = [v_1 \ v_2]$ is a probability vector, show that VA is also a probability vector.

6 L2 5 1

- b) A software engineer goes to his workplace every day either by bike or by car. He never goes by bike on two consecutive days. If he goes by car on a day then he is equally likely to go by car or bike on the next day. Find the transition matrix for the chain of the mode of transport he uses. If car is used on the first day of the week, find the probability that after four days:

- i) bike is used ii) car is used.

6 L3 5 1

- c) Show that the Markov Chain with transition matrix P is irreducible.

$$P = \begin{pmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

4 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
